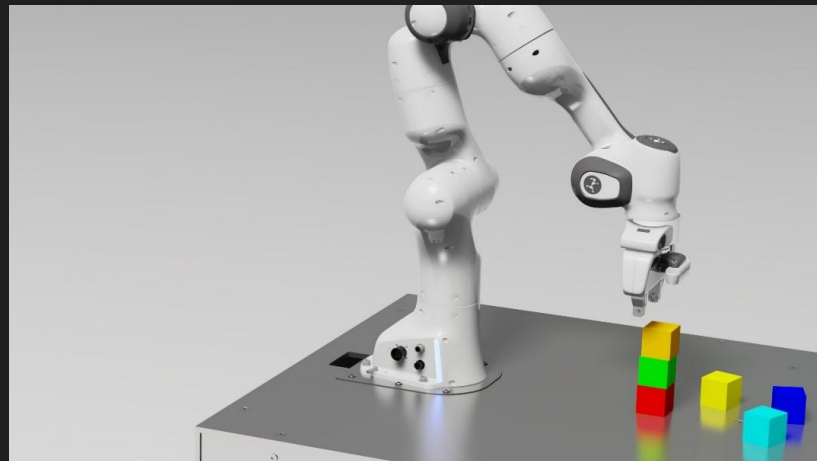
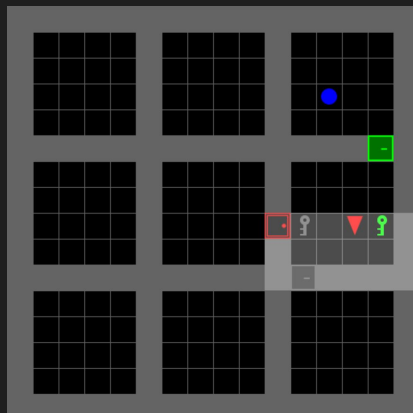


From Pixels to Plans: Training VLA Agents via Curriculum Reinforcement Learning



Presenter: Carolina Li and Peiyao Tao
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Course: CS7180 Advanced Perception
Institution: Northeastern University

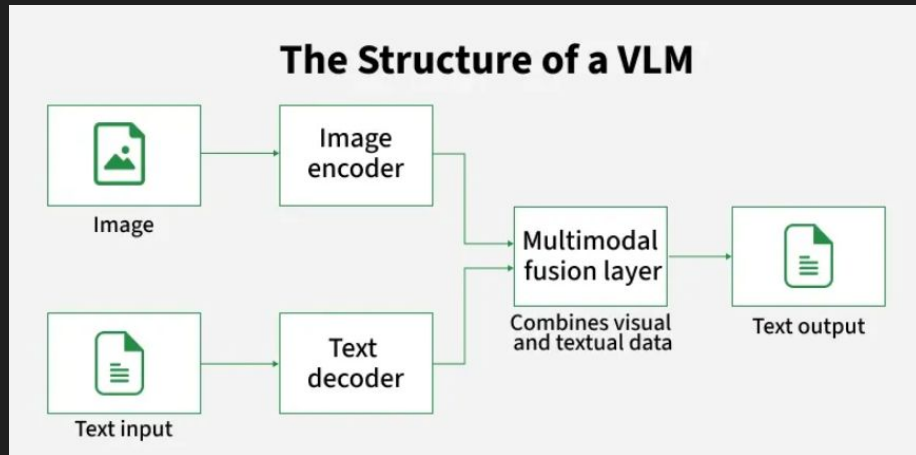
Motivation & Goal

Why this project?

- Vision is essential for embodied intelligence
- RL on raw pixels is difficult (sparse rewards, partial observability)
- Build a **progressive curriculum** from simple → complex
- Establish a **baseline** before adding Vision–Language Models (CLIP, BLIP, VLA agents)

Background: Vision Language Model & Reinforcement Learning

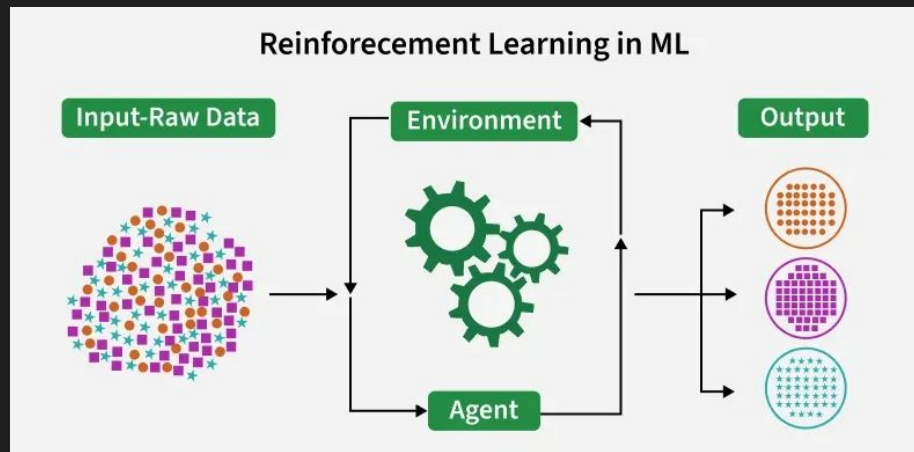
- Jointly model **visual perception** and **natural language understanding**
- Learn aligned representations from **image–text pairs**
- Enable multimodal tasks:
 - Image captioning
 - Visual Question Answering (VQA)
 - Cross-modal retrieval
- Common backbone:
 - Vision encoder (CNN / ViT)
 - Language encoder (Transformer)
 - Shared embedding space



Examples: CLIP, ALIGN, ViLT

Background: Reinforcement Learning

- Learn decision-making by **trial and error**
- Core loop:
 - **State/Observation** → **Action** → **Reward** → **Next state**
- Goal: maximize **expected cumulative reward**
- Key concepts:
 - **Policy**: how the agent acts
 - **Reward**: feedback signal
 - **Value**: long-term usefulness
- RL is commonly modeled as an **(PO)MDP**
(*partial observability is typical in robotics and MiniGrid*)



In this project: PPO learns a policy from pixels under sparse rewards.

The Vision-Language-Action (VLA) Architecture

Visual Encoder:

Custom 3-Layer CNN to extract edges, colors, and shapes.

Spatial Abstraction:

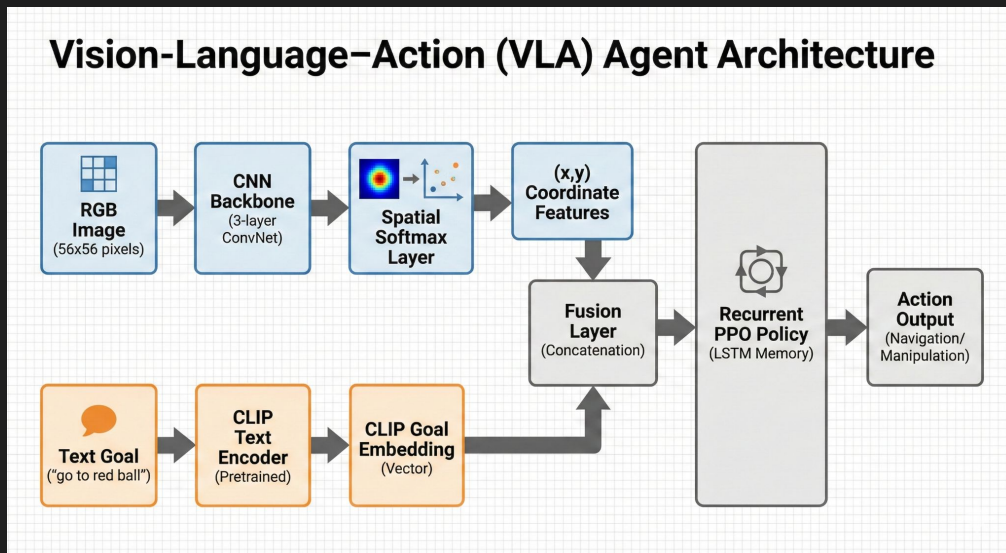
A Spatial Softmax layer converts abstract feature maps into precise (x,y) coordinates.

Language Grounding:

A frozen CLIP Text Encoder embeds the goal (e.g., "find the ball") into a vector.

Fusion:

Visual coordinates and Goal vectors are concatenated before entering the Policy Network (Recurrent PPO).



Solving "Physics Blindness" with Spatial Softmax

- **The Problem with Standard CNNs:**
 - Standard networks use Global Average Pooling, which destroys spatial information. It tells the agent "There is a key," but not where it is.
- **Our Solution:** Spatial Softmax
 - Forces the network to output heat maps representing object locations.
 - Calculates the expected center of mass (x,y) for these features.
- **Result:**
 - The agent effectively learns an internal coordinate system from raw pixels, enabling it to navigate to specific sub-goals.

Object Permanence & The LSTM

- **The Challenge:**
 - The goal is often inside a locked room. If the agent turns away, the object vanishes from the pixel input.
- **The Solution:**
 - Recurrent PPO (LSTM Memory).
- **Perception over Time:**
 - The CNN extracts visual features from the current frame.
 - The LSTM integrates these features into a hidden "Belief State."
 - Outcome: The agent "remembers" the location of a box it saw 50 steps ago, effectively solving the Object Permanence problem.

Phase Overview

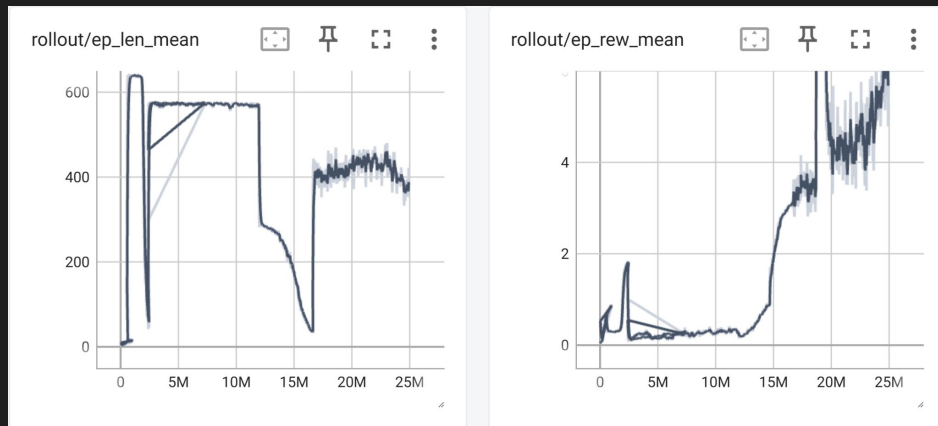
- Phase 1 — GoToObject
- Phase 2 — DoorKey Task
- Phase 3 — ObstructedMaze-1d1h
- Phase 4 — ObstructedMaze-2d1h: Final Maze

Each phase increases:

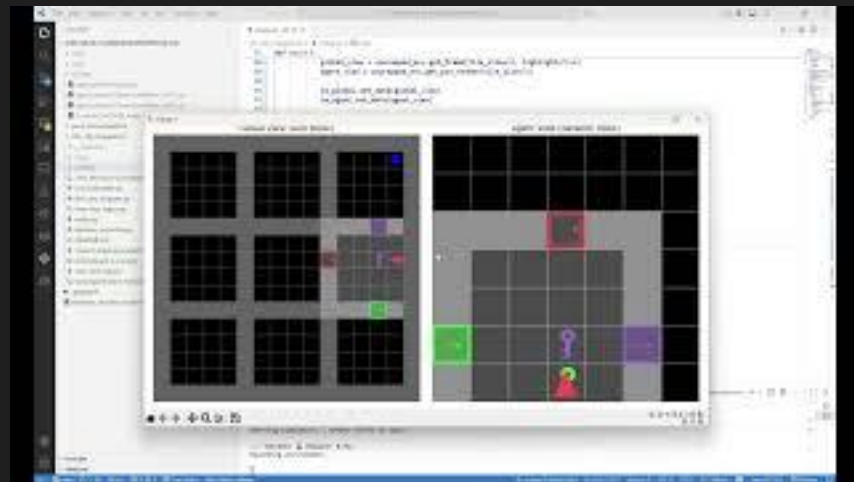
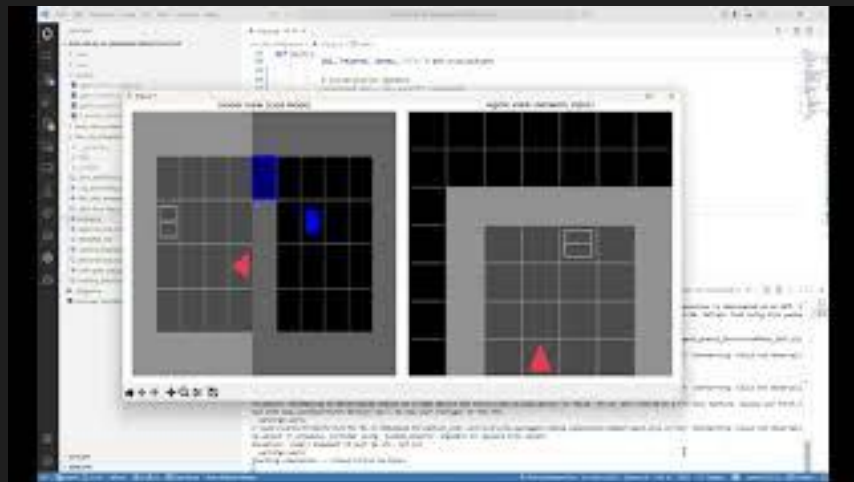
- Environment size
- Required reasoning
- Reward hierarchy
- Number of steps to succeed

Results

- **Metric:**
 - Achieved ~60% Training Success Rate (Rollout Reward 6.37) on the hardest task.
- **Inference:**
 - The high success rate proves the Vision Encoder successfully locates sub-goals and the final goal.
 - The agent successfully ignores "distractor" objects (walls, incorrect doors).
- **Generalization:**
 - The perception module handles Randomized Maps (Procedural Generation) without retraining, proving it isn't memorizing layouts.



Demo



Conclusions

- **Achievement:**
 - Successfully trained a pixel-based agent to solve multi-stage reasoning puzzles.
- **Key Driver:**
 - The combination of Spatial Softmax (for precise localization) and CLIP (for semantic goal understanding).
- **Takeaway:**
 - We bridged the gap between high-dimensional visual inputs and low-level discrete actions without using symbolic cheat sheets.

Discussions and Future Works

- 📌 Sparse pixel-based RL is extremely hard
- 📌 Reward shaping is essential
- 📌 Hierarchical tasks expose PPO's limitations
- 📌 Phase progression builds a curriculum for VLM-enhanced RL

Future upgrades:

- Use CLIP features (instead of raw pixels)
- Use language goals (“find the blue ball”)
- Add memory modules or decision transformers
- Integrate action-chunking / hierarchical policies

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